

Fermion: TCS3200 Color Sensor



SKU: SEN0101

The Gravity: TCS3200 Color Sensor is a comprehensive color detection tool equipped with a TAOS TCS3200 RGB sensor chip and white LEDs, designed for Arduino applications. It is capable of detecting a wide range of visible colors, making it ideal for various uses such as test strip reading, color sorting, ambient light sensing, calibration, and color matching. The sensor consists of an array of photo detectors with red, green, blue, or no filters, ensuring balanced color detection across the array. An internal oscillator generates a square-wave output, providing a frequency proportional to the intensity of the detected color, thus enabling precise color measurement.

 [Buy now](#)

[Docs](#) [Tech Specs](#)

Specification

Parameter	Description
Single-Supply Operation	2.7V to 5.5V
High-Resolution Conversion	Light Intensity to Frequency
Programmable Features	Color and Full-Scale Output Frequency
Power Management	Power Down Feature



[Store](#) ▾ [Community](#)

[Forum](#)

[Wiki](#)

[Blog](#)

[Learn](#)



[Global Site](#)

[\\$USD](#)



 Search

[← Sensors](#)

**Light & Imaging
Sensors**

ESP32-S3 AI Camera Module(SKU: DFR1154)

Gravity: IR Positioning Camera For Arduino (SKU: SEN0158)

RS485 Photosynthetically Active Radiation (PAR) Sensor (400-700nm)(SKU: SEN0641)

HUSKYLENS GC2093 Microscope Imaging Module(SKU: SEN0683)

AI Binocular Face, Palm Vein & QR Code Recognition Sensor(SKU: SEN0677)

MS-A075V RGB-D TOF Camera Module (SKU: SEN0583)

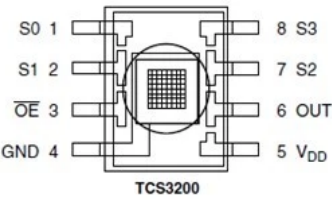
RS485 Waterproof Ambient Light Sensor (0-200,000 Lux)(SKU: SEN0644)

Gravity: Analog Ambient Light Sensor(SKU: DFR0026)

RS485 Photoelectric Solar Radiation Sensor

Parameter	Description
Communication	Communicates Directly to Microcontroller
S0~S1	Output frequency scaling selection inputs
S2~S3	Photodiode type selection inputs
OUT Pin	Output frequency
OE Pin	Output frequency enable pin (active low), can be impending when using
Additional Feature	Support LED lamp light supplement control
Size	28.4x28.4mm

Pinout



Pin Name	I/O	DESCRIPTION
GND(4)	GND	Power supply ground. All voltages are referenced to GND
OE(3)	I	Enable for fo (active low)
OUT	O	Output frequency (of)
S0, S1 (1, 2)	I	Output frequency scaling selection inputs
S2, S3 (7, 8)	I	Photodiode type selection inputs
VDD (5)	VDD	Supply voltage

Was this article helpful?



Information

- About Us
- Warranty
- Terms & Conditions
- Shipping
- Payment
- FAQ

Customer Service My Account

- DFRobot Distributors
- Contact Us
- Site Map
- Affiliates
- Specials



Sign up for exclusive offers!

Your email address >

Like us on

